Name: ____________________ Date: __________

Induction Set 1 [67 marks]

1. It is given that $2\cos A \sin B \equiv \sin(A+B) - \sin(A-B)$. (Do **not** prove this identity.)

Using mathematical induction and the above identity, prove that $\sum_{r=1}^{n} \cos(2r-1)\theta = \frac{\sin 2n\theta}{2\sin\theta}$ for $n \in \mathbb{Z}^+$.

$P_n : \sum_{r=1}^{n} (\cos((2r-1)\theta)) = \frac{\sin 2n\theta}{2\sin\theta}$ for $n \geq 1$

$P_1 : \sum_{r=1}^{1} \cos((2(1)-1)\theta) = \cos\theta$

$\frac{\sin 2\theta}{2\sin\theta} = \frac{2\sin\theta\cos\theta}{2\sin\theta} = \cos\theta$

Assume

$P_k : \sum_{r=1}^{k} \cos((2r-1)\theta) = \frac{\sin 2k\theta}{2\sin\theta}$

$P_{k+1} : \sum_{r=1}^{k+1} \cos((2r-1)\theta)$

$= \frac{\sin 2k\theta}{2\sin\theta} + \frac{\cos((2k+1)\theta)}{1}$

A B

$= \frac{\sin 2k\theta}{2\sin\theta} + \frac{2\cos((2k+1)\theta)\sin\theta}{2\sin\theta}$

$= \frac{\sin 2k\theta + \sin((2k+1)\theta+\theta) - \sin((2k+1)\theta-\theta)}{2\sin\theta}$

$= \frac{\cancel{\sin 2k\theta} + \sin(2(k+1)\theta) - \cancel{\sin 2k\theta}}{2\sin\theta}$ ✓

$\therefore P_n : \sum_{r=1}^{n} \cos((2r-1)\theta) = \frac{\sin 2n\theta}{2\sin\theta}$ for $n \geq 1$ by induction

[8]

2.

[Maximum mark: 7]

Using mathematical induction and the definition ${}^{n}C^{r} = \frac{n!}{r!(n-r)!}$, prove that $\sum_{r=1}^{n} {}^{r}C^{1} = {}^{n+1}C_2$ for all $n \in \mathbb{Z}^+$.

$P_n : \sum_{r=1}^{n} {}_{r}C_1 = {}_{n+1}C_2$ for $n \geq 1$

$P_1 : \sum_{r=1}^{1} {}_{r}C_1 = {}_{1}C_1 = 1$

${}_{2}C_2 = 1$ ✓

Assume

$P_k : \sum_{r=1}^{k} {}_{r}C_1 = {}_{k+1}C_2$

$P_{k+1} : \sum_{r=1}^{k+1} {}_{r}C_1 = {}_{k+1}C_2 + {}_{k+1}C_1$

$= \frac{(k+1)!}{2!(k-1)!} + \frac{(k+1)!}{k!}$

$= \frac{(k+1)k}{2} + k+1 = \frac{(k+1)k + 2(k+1)}{2}$

$= \frac{(k+1)(k+2)}{2} = {}_{k+2}C_2$ ✓

$\therefore P_n : \sum_{r=1}^{n} {}_{r}C_1 = {}_{n+1}C_2$ for $n \geq 1$ by induction hardest one

3. (a) Let $f(x) = (1-ax)^{-\frac{1}{2}}$, where $ax < 1$, $a \neq 0$. The n^{th} derivative of $f(x)$ is denoted by $f^{(n)}(x)$, $n \in \mathbb{Z}^+$. Prove by induction that $f^{(n)}(x) = \frac{a^n(2n-1)!(1-ax)^{-\frac{2n+1}{2}}}{2^{2n-1}(n-1)!}$, $n \in \mathbb{Z}^+$.

$$P_n : f^n(x) = \frac{a^n(2n-1)!\,(1-ax)^{-\left(\frac{2n+1}{2}\right)}}{2^{2n-1}(n-1)!}$$

$$P_1 : f'(x) = -\tfrac{1}{2}\cdot(-a)(1-ax)^{-\frac{3}{2}}$$

$$= \frac{a^1(1!)(1-ax)^{-\frac{3}{2}}}{2(0!)} \checkmark$$

Constant

$$\text{Assume } P_k : f^k(x) = \boxed{\frac{a^k(2k-1)!\,(1-ax)^{-\left(\frac{2k+1}{2}\right)}}{2^{2k-1}(k-1)!}}$$

$$P_{k+1} : f^{k+1} = \boxed{\frac{a^k(2k-1)!}{2^{2k-1}(k-1)!}} \cdot \left[-a\cdot\left(-\left(\frac{2k+1}{2}\right)\right)\right] \cdot [1-ax]^{-\left(\frac{2(k+1)+1}{2}\right)}$$

Chain Rule

Power Rule

$$= \left[\frac{a^{k+1}}{2^{2k}}\right] \cdot \left[\frac{(2k-1)!\,(2k+1)}{(k-1)!} \cdot \frac{2k}{2k}\right] \cdot [1-ax]^{-\left(\frac{2(k+1)+1}{2}\right)}$$

I did this to create a factorial

$$\left[\frac{a^{k+1}}{2^{2(k+1)-1}}\right] \cdot \left[\frac{(2k+1)!}{k!}\right] [1-ax]^{-\left(\frac{2(k+1)+1}{2}\right)} \checkmark$$

$\therefore P_n$ is true for $n \geq 1$ by induction

[8]

(b) By using part (a) or otherwise, show that the Maclaurin series for $f(x) = (1-ax)^{-\frac{1}{2}}$ up to and including the x^2 term is $1 + \frac{1}{2}ax + \frac{3}{8}a^2x^2$.

Binomial Extension Formula

$$(1-ax)^{-\frac{1}{2}} \simeq 1 + \left(-\tfrac{1}{2}\right)(-ax) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)(-ax)^2}{2!}$$

$$\approx 1 + \tfrac{1}{2}ax + \tfrac{3}{8}a^2x^2 \checkmark$$

[2]

(c) Hence, show that $(1-2x)^{-\frac{1}{2}}(1-4x)^{-\frac{1}{2}} \approx \frac{2+6x+19x^2}{2}$.

$$(1-2x)^{-\frac{1}{2}}(1-4x)^{-\frac{1}{2}}$$

$$\approx \left(1+x+\tfrac{3}{2}x^2\right)\left(1+2x+6x^2\right)$$

$$\approx 1+x+2x+2x^2+\tfrac{3}{2}x^2+6x^2$$

$$\approx 1+3x+\tfrac{19}{2}x^2$$

$$\approx \frac{2+6x+19x^2}{2} \checkmark$$

[4]

(d) Given that the series expansion for $(1-ax)^{-\frac{1}{2}}$ is convergent for $|ax| < 1$, state the restriction which must be placed on x for the approximation $(1-2x)^{-\frac{1}{2}}(1-4x)^{-\frac{1}{2}} \approx \frac{2+6x+19x^2}{2}$ to be valid.

$$-\frac{1}{4} < x < \frac{1}{4}$$

[1]

(e) Use $x = \frac{1}{10}$ to determine an approximate value for $\sqrt{3}$. Give your answer in the form $\frac{c}{d}$, where $c,\ d \in \mathbb{Z}^+$.

$$\frac{1}{\sqrt{\frac{4}{5}}} \cdot \frac{1}{\sqrt{\frac{3}{5}}} \approx \frac{2+\frac{3}{5}+\frac{19}{100}}{2}$$

$$\frac{5}{2\sqrt{3}} \approx \frac{279}{200}$$

[5]

$$\frac{2\sqrt{3}}{5} \approx \frac{200}{279}$$

$$\sqrt{3} \approx \frac{500}{279}$$

4. Prove by mathematical induction that $5^{2n} - 2^{3n}$ is divisible by 17 for all $n \in \mathbb{Z}^+$.

$P_n: 5^{2n} - 2^{3n} = 17A$ for $A \in \mathbb{Z}$ and $n \geq 1$

$P_1: 5^2 - 2^3 = 25 - 8 = 17 = 17(1)$ ✓

Assume $P_K: 5^{2K} - 2^{3K} = 17B$ for $B \in \mathbb{Z}$

$P_{K+1}: 5^{2(K+1)} - 2^{3(K+1)}$

$= 25(17B + 2^{3K}) - 8(2^{3K})$

$= 25(17B) + 17(2^{3K})$

$= 17(25B + 2^{3K})$ ✓

$\therefore P_n: 5^{2n} - 2^{3n} = 17A$ for $A \in \mathbb{Z}$ for $n \geq 1$ by induction

[7]

5. Use mathematical induction to prove that $\sum_{r=1}^{n} \frac{r}{(r+1)!} = 1 - \frac{1}{(n+1)!}$ for all integers $n \geq 1$.

$P_n: \sum_{r=1}^{n} \frac{r}{(r+1)!} = 1 - \frac{1}{(n+1)!}$ for $n \geq 1$

$P_1: \sum_{r=1}^{1} \frac{r}{(r+1)!} = \frac{1}{2!} = \frac{1}{2}$

$1 - \frac{1}{(1+1)!} = \frac{1}{2}$ ✓

Assume $P_K: \sum_{r=1}^{K} \frac{r}{(r+1)!} = 1 - \frac{1}{(K+1)!}$

$P_{K+1}: \sum_{r=1}^{K+1} \frac{r}{(r+1)!} = 1 - \frac{1}{(K+1)!} + \frac{K+1}{(K+2)!}$

$= 1 - \frac{(K+2)! - (K+1)(K+1)!}{(K+1)!(K+2)!}$

$= 1 - \frac{\cancel{(K+1)!}[\cancel{K}+2-\cancel{K}-1]}{\cancel{(K+1)!}(K+2)!} = 1 - \frac{1}{(K+2)!}$ ✓

$\therefore P_n: \sum_{r=1}^{n} \frac{r}{(r+1)!} = 1 - \frac{1}{(n+1)!}$ for $n \geq 1$ by induction

[7]

6. Let z_n be the complex number defined as $z_n = (n^2 + n + 1) + i$ for $n \in \mathbb{N}$.

(a.i) Find $\arg(z_0)$.

$$z_0 = 1 + i \quad \arg(z_0) = \frac{\pi}{4}$$

[2]

(a.ii) Write down an expression for $\arg(z_n)$ in terms of n.

[1]

$$\arctan\left(\frac{1}{n^2+n+1}\right)$$

Let $w_n = z_0 z_1 z_2 z_3 \ldots z_{n-1} z_n$ for $n \in \mathbb{N}$.

(b.i) Show that $\arctan(a) + \arctan(b) = \arctan\left(\frac{a+b}{1-ab}\right)$ for $a, b \in \mathbb{R}^+$, $ab < 1$.

$$\tan\left(\arctan(a) + \arctan(b)\right) = \frac{\tan(\arctan(a)) + \tan(\arctan(b))}{1 - \tan(\arctan(a))\tan(\arctan(b))}$$

angles

Use compound identity

$$\tan(\arctan a + \arctan b) = \frac{a+b}{1-ab}$$

[2]

$\therefore \arctan(a) + \arctan(b)$

$$= \arctan\left(\frac{a+b}{1-ab}\right)$$

(b.ii) Hence or otherwise, show that $\arg(w_1) = \arctan(2)$.

$$\arg(w_1) = \arg(z_0 \cdot z_1)$$

$$= \arg(z_0) + \arg(z_1)$$

(De Moivre's Theorem)

$$= \arctan(1) + \arctan\left(\tfrac{1}{3}\right) = \arctan\left(\frac{1+\frac{1}{3}}{1-\frac{1}{3}}\right) = \arctan(2) \checkmark$$

[3]

(c) Prove by mathematical induction that $\arg(w_n) = \arctan(n+1)$ for $n \in \mathbb{N}$. This one is hard

[10]

$P_n : \arg(w_n) = \arctan(n+1)$ for $n \geq 1$

$P_1 : \arg(w_1) = \arctan(2)$

Proven above

$\therefore P_n : \arg(w_n) = \arctan(n+1)$ for $n \geq 1$ by induction

Assume

$P_k : \arg(w_k) = \arctan(k+1)$

$P_{k+1} : \arg(w_{k+1}) = \arg(z_{k+1}) + \arg(w_k)$

De Moivre's Theorem

$$= \arctan\left(\frac{1}{(k+1)^2 + k + 2}\right) + \arctan(k+1)$$

$$= \arctan\left(\frac{\frac{1}{k^2+3k+3} + k+1}{1 - \frac{k+1}{k^2+3k+3}}\right) = \arctan\left(\frac{(k+1)(k^2+3k+3)+1}{k^2+3k+3-k-1}\right) = \arctan\left(\frac{k^3+4k^2+6k+4}{k^2+2k+2}\right)$$

$$= \arctan\left(\frac{(k+2)(k^2+2k+2)}{k^2+2k+2}\right)$$

$$= \arctan(k+2) \checkmark$$

$$\begin{array}{r|rrrr} -2 & 1 & 4 & 6 & 4 \\ & & -2 & -4 & -4 \\ \hline & 1 & 2 & 2 & \boxed{0} \end{array}$$